

## 01 CONDENSED LECTURE NOTES

## Metrics and the geometry of balls

Source: I.1 and II.1-II.2.1, original pp. 3-10

Bracketed labels retain the original numbering. Proofs are reduced to occasional proof ideas; hypotheses and key distinctions are retained.

**Metric and metric space.** [II.1] A metric on a nonempty set  $X$  is a function  $d : X \times X \rightarrow \mathbb{R}$  such that, for all  $x, y, z \in X$ , Ex1 q1,5,6

- **Positive definite:**  $d(x, y) \geq 0$ , and  $d(x, y) = 0 \iff x = y$ .
- **Symmetry:**  $d(x, y) = d(y, x)$ .
- **$\triangle$  inequality:**  $d(x, z) \leq d(x, y) + d(y, z)$ .

The pair  $(X, d)$  is a *metric space*. Its points need not be numbers: they may be functions or other objects. A nonempty subset  $A \subseteq X$ , with  $d$  restricted to  $A \times A$ , is a *metric subspace*.

## Standard examples [I.1; II.2-II.9]

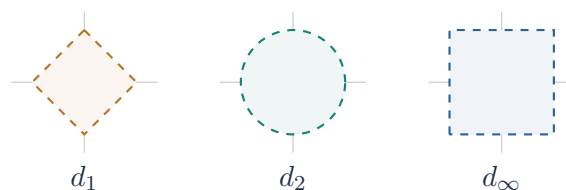
| Space                             | Metric                                                                                                                                              |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbb{R}$ or $\mathbb{C}$      | $d(x, y) =  x - y $                                                                                                                                 |
| $\mathbb{R}^n$ or $\mathbb{C}^n$  | $d_1(x, y) = \sum_{j=1}^n  x_j - y_j $<br>$d_2(x, y) = (\sum_{j=1}^n  x_j - y_j ^2)^{1/2}$<br>$d_\infty(x, y) = \max_{1 \leq j \leq n}  x_j - y_j $ |
| $B(S)$ , bounded functions on $S$ | $d_\infty(f, g) = \sup_{s \in S}  f(s) - g(s) $                                                                                                     |
| $C[a, b]$                         | The same supremum metric, restricted to continuous functions                                                                                        |
| $C[0, 1]$                         | Integral metric: $d_1(f, g) = \int_0^1  f(x) - g(x)  dx$                                                                                            |
| Any nonempty set $X$              | Discrete metric: $d(x, y) = 1$ if $x \neq y$ , and 0 otherwise                                                                                      |

**Open and closed balls.** [II.18] For  $x \in X$  and  $r > 0$ ,

$$B_r(x) = \{y \in X : d(x, y) < r\}, \quad B_r[x] = \{y \in X : d(x, y) \leq r\}.$$

In  $\mathbb{R}$ , these are  $(x - r, x + r)$  and  $[x - r, x + r]$ . In a subspace  $A$ ,  $B_r^A(x) = A \cap B_r^X(x)$ . In the discrete metric,

$$B_r(x) = \begin{cases} \{x\}, & 0 < r \leq 1, \\ X, & r > 1, \end{cases} \quad B_r[x] = \begin{cases} \{x\}, & 0 < r < 1, \\ X, & r \geq 1. \end{cases}$$



Open unit balls in  $\mathbb{R}^2$ : diamond, disc and square. Their boundaries are excluded.

**Distance to a set.** [II.10] For nonempty  $A \subseteq X$ , define  $d(x, A) = \inf_{a \in A} d(x, a)$ .

Intuitively, this is the minimum distance between  $x$  and a point of  $A$ . A closest point of  $A$  need not exist, so this “minimum” need not be attained; this is why the definition uses an infimum. In particular,  $d(x, A) = 0$  does not necessarily imply  $x \in A$ . Ex10 q5

## 02 CONDENSED LECTURE NOTES

## Convergence and equivalent metrics

Source: I.1, II.1.1 and II.22, original pp. 3-10

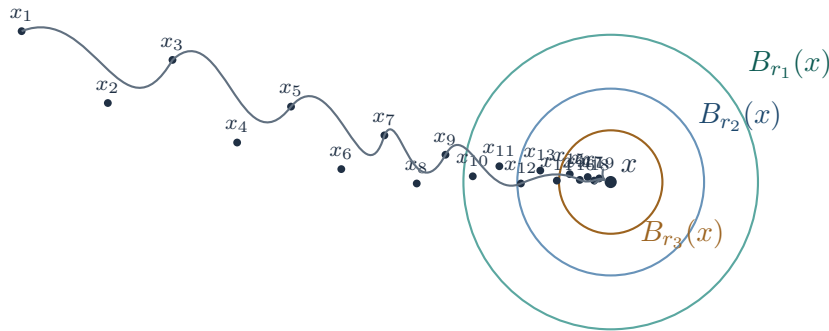
**Convergence in a metric space.** [II.11] A sequence  $(x_n)$  in  $(X, d)$  converges to  $x \in X$  if

$$\forall \varepsilon > 0 \exists N_\varepsilon \forall n > N_\varepsilon : d(x_n, x) < \varepsilon.$$

Equivalently, the real sequence  $d(x_n, x)$  tends to 0. The index may depend on  $\varepsilon$ ; every sufficiently late term must satisfy the inequality.**Uniqueness and neighbourhood criterion.** [II.12, II.26]

- **Uniqueness.** A sequence in a metric space has at most one limit.
- **Neighbourhood criterion.**

$$x_n \rightarrow x \iff \begin{array}{c} \text{every open neighbourhood of } x \\ \text{contains} \\ \text{all sufficiently late } x_n \end{array} \iff \forall r > 0, B_r(x) \text{ contains} \\ \text{all sufficiently late } x_n.$$

*Proof idea for uniqueness:* if  $x_n \rightarrow x$  and  $x_n \rightarrow y$ , then  $d(x, y) \leq d(x, x_n) + d(x_n, y) \rightarrow 0$ .**Intuition.** However small a ball around  $x$  is, the tail of the sequence eventually lies inside it. You can think of the sequence as producing infinitely many points in arbitrarily small balls around  $x$ ; the points may wiggle as they approach  $x$ , but they get closer together and denser and denser near the limit.**Equivalent metrics.** [II.14] Metrics  $d$  and  $\rho$  on the same set  $X$  are equivalent if they have exactly the same convergent sequences *with the same limits*: Ex1 q7; Ex2 q1(i-ii)

$$x_n \xrightarrow{d} x \iff x_n \xrightarrow{\rho} x.$$

**Ball criterion for equivalence** [II.22]. For every  $x \in X$ :

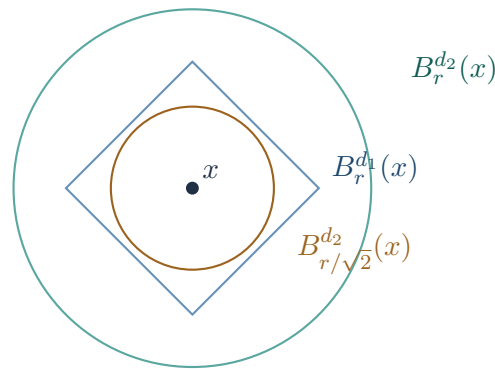
- every  $\rho$ -ball about  $x$  contains a  $d$ -ball about  $x$ ;
- every  $d$ -ball about  $x$  contains a  $\rho$ -ball about  $x$ .

Hence  $d$  and  $\rho$  induce the same open sets.A useful sufficient condition is  $c d \leq \rho \leq C d$ . Then

$$B_{r/C}^d(x) \subseteq B_r^\rho(x) \subseteq B_{r/c}^d(x),$$

so a scaled  $d$ -ball always fits inside the  $\rho$ -ball, and another scaled  $d$ -ball contains the  $\rho$ -ball. For the concrete example  $d = d_2$  and  $\rho = d_1$  on  $\mathbb{R}^2$ ,

$$d_2 \leq d_1 \leq \sqrt{2} d_2 \implies B_{r/\sqrt{2}}^{d_2}(x) \subseteq B_r^{d_1}(x) \subseteq B_r^{d_2}(x).$$



Thus every  $d_1$ -ball contains a smaller  $d_2$ -ball and is itself contained in a  $d_2$ -ball. Since this can be done around every point and at every scale, the ball criterion applies:  $d_1$  and  $d_2$  induce the same open sets, hence the two metrics are equivalent.

**Uniform and pointwise convergence.** [II.15-II.17] For  $f_n, f \in B(S)$ :

- **Uniform.**  $f_n \rightarrow f$  uniformly iff  $\|f_n - f\|_\infty \rightarrow 0$ , equivalently

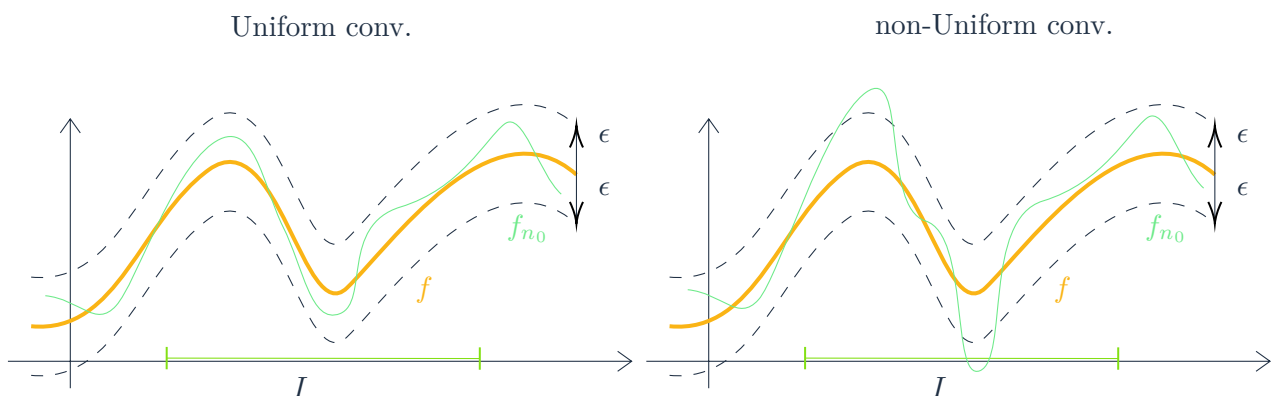
$$\forall \varepsilon > 0 \exists N_\varepsilon \forall n > N_\varepsilon \forall x \in S : |f_n(x) - f(x)| < \varepsilon.$$

- **Pointwise.**  $f_n \rightarrow f$  pointwise iff for each fixed  $x \in S$ ,

$$\forall \varepsilon > 0 \exists N_{\varepsilon, x} \forall n > N_{\varepsilon, x} : |f_n(x) - f(x)| < \varepsilon.$$

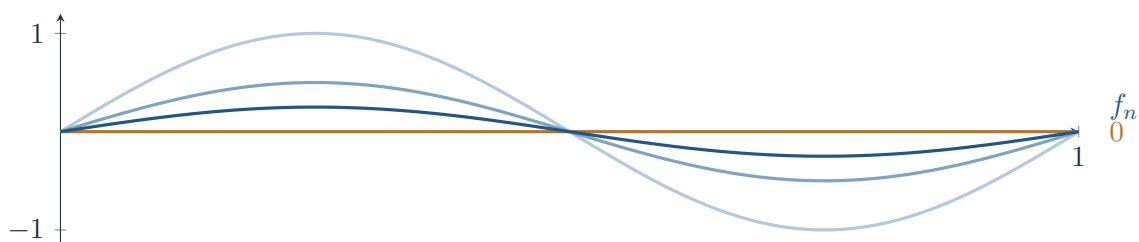
The key difference is that for pointwise convergence the index  $N$  may depend on both  $x$  and  $\varepsilon$ , while for uniform convergence  $N$  depends only on  $\varepsilon$ . Uniform convergence implies pointwise convergence, but not conversely.

**Intuition.** The only difference is where the phrase “for all  $x \in S$ ” sits. In uniform convergence, once  $n \geq N(\varepsilon)$ , the entire graph of  $f_n$  must fit inside the  $\varepsilon$ -belt around  $f$ .



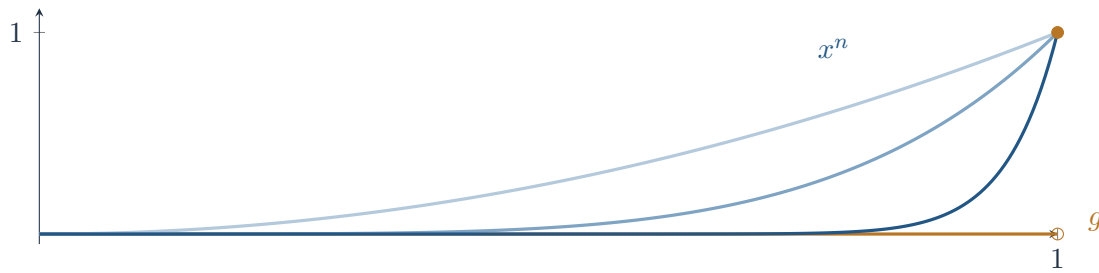
## Examples that distinguish the notions

### 1. A genuinely uniform example.



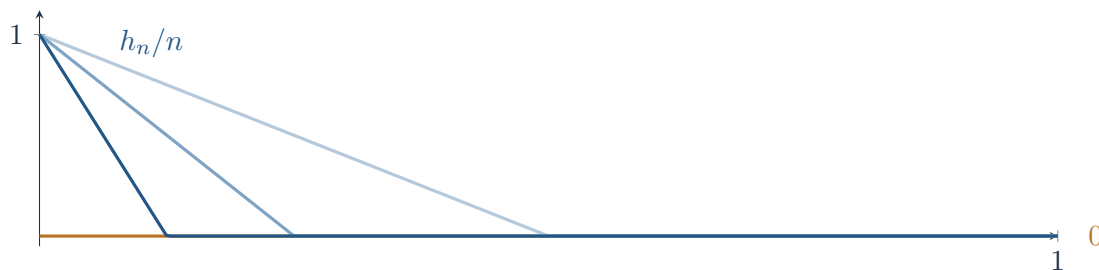
Here the whole graph shrinks vertically toward 0: for  $f_n(x) = \sin(2\pi x)/n$  we have  $\|f_n\|_\infty = 1/n \rightarrow 0$ , so the convergence is uniform.

## 2. Pointwise but not uniform.



The graphs  $g_n(x) = x^n$  converge to 0 for every  $x < 1$ , but stay equal to 1 at  $x = 1$ . Hence the pointwise limit is discontinuous and the convergence is not uniform.

## 3. Convergence in $d_1$ but not in $d_\infty$ .



These spikes become narrower and narrower, so the area under them tends to 0 and hence  $d_1(h_n/n, 0) \rightarrow 0$ . But their height stays equal to 1, so  $d_\infty(h_n/n, 0) = 1$  for all  $n$ .

**Keep in mind.** Equivalent metrics preserve convergence and open sets, but need not preserve Cauchy sequences or completeness.

## 03 CONDENSED LECTURE NOTES

# Open sets, closed sets and closure

Source: II.2.2-II.2.3, original pp. 10-15

**Open set and neighbourhood.** [II.23] A set  $U \subseteq X$  is open if every  $x \in U$  has some  $r > 0$  with  $B_r(x) \subseteq U$ . The notes use *neighbourhood* to mean an *open* set containing the point.

**Intuition.** At every point of an open set we can fit a sufficiently small open ball which still lies completely inside the set.

**Consequence: open balls are open.** If  $y \in B_r(x)$ , then  $r - d(x, y) > 0$  and

$$B_{r-d(x,y)}(y) \subseteq B_r(x),$$

so we can fit a small ball around every point  $y$  of  $B_r(x)$ .

**Basic open-set results.** [II.25, II.27, II.33]

- **Trivial open sets.**  $\emptyset$  and  $X$  are open.
- **Arbitrary unions.** An arbitrary union of open sets is open.
- **Finite intersections.** A finite intersection of open sets is open.
- **Union-of-balls characterisation.** A set is open if and only if it is a union of open balls.

An infinite intersection need not be open:  $\bigcap_{n \geq 1} (-1/n, 1/n) = \{0\}$  in  $\mathbb{R}$ .

**Relative openness.** [II.34] For a subspace  $A \subseteq X$ , a set  $U \subseteq A$  is open in  $A$  if and only if  $U = A \cap O$  for some open  $O \subseteq X$ . For example,  $[0, 1/2)$  is open in  $[0, 1]$ , although not in  $\mathbb{R}$ .

**Interior.** [II.35]

$$\text{int}(A) = \{x \in A : \exists r > 0, B_r(x) \subseteq A\} = \bigcup_{U \subseteq A, U \text{ open}} U.$$

Ex3  
q1,3

It is the largest open subset of  $A$ ;  $A$  is open if and only if  $A = \text{int}(A)$ .

**Limit point (accumulation point).** [II.36, II.38] A point  $x \in X$  is a limit point of  $A$  if every  $B_r(x)$  meets  $A \setminus \{x\}$ . Write  $A'$  for the set of limit points. Equivalently, there is a sequence  $a_n \in A \setminus \{x\}$  with  $a_n \rightarrow x$ . A limit point need not belong to  $A$ .

**Closed set: equivalent tests in metric spaces.** [II.39-II.42] For  $A \subseteq X$ , the following are equivalent: Ex3

- (i)  $A$  contains every limit point of  $A$ :  $A' \subseteq A$ ;
- (ii) every sequence in  $A$  that converges in  $X$  has its limit in  $A$ ;
- (iii)  $X \setminus A$  is open.

q4,8

Closed balls are closed. Arbitrary intersections and finite unions of closed sets are closed;  $\emptyset$  and  $X$  are both open and closed. A set may be open only, closed only, both, or neither.

**Closure and its equivalent descriptions** [II.45-II.49]. The closure  $\overline{A}$  is the smallest closed subset of  $X$  containing  $A$ :

$$\overline{A} = \bigcap_{F \supseteq A, F \text{ closed}} F = A \cup A' = \{x : d(x, A) = 0\} \quad (A \neq \emptyset).$$

Also,  $x \in \overline{A}$  if and only if every open neighbourhood of  $x$  meets  $A$ ; in a metric space this is equivalent to  $a_n \rightarrow x$  for some sequence  $(a_n) \subseteq A$ .

**Example.** For  $A = [0, 1) \cup \{2\} \subset \mathbb{R}$ ,

$$\text{int}(A) = (0, 1), \quad A' = [0, 1], \quad \overline{A} = [0, 1] \cup \{2\}.$$

The isolated point 2 is in the closure, but not in  $A'$ .

**Keep in mind.**  $\overline{B_r(x)} \subseteq B_r[x]$ , but equality can fail. In a discrete space with at least two points,  $\overline{B_1(x)} = \{x\}$  while  $B_1[x] = X$ . Openness and closedness always refer to a specified ambient space.

#### 04 CONDENSED LECTURE NOTES

## Topological spaces and Hausdorff separation

Source: III, original pp. 16-19

If a pattern repeats sufficiently often in mathematics, it can become the definition of a more general concept. The properties of open sets above now become the defining properties of a topology, which generalizes the concept of a metric space.

**Topology.** [III.1-III.2] A topology  $\mathcal{T}$  on a set  $X$  is a collection of subsets of  $X$  satisfying:

Ex4  
q2,6

- **Triviality.**  $\emptyset, X \in \mathcal{T}$ .
- **Arbitrary unions.** The union of any family of members of  $\mathcal{T}$  belongs to  $\mathcal{T}$ .
- **Finite intersections.** The intersection of finitely many members of  $\mathcal{T}$  belongs to  $\mathcal{T}$ .

The pair  $(X, \mathcal{T})$  is a topological space; its members  $U \in \mathcal{T}$  are the open sets. A metric generates a topology by declaring its metric-open sets open.

#### Examples and comparison [III.3-III.10]

- **Discrete topology:**  $\mathcal{T} = \mathcal{P}(X)$ , so every subset is open.
- **Trivial (indiscrete) topology:**  $\mathcal{T} = \{\emptyset, X\}$ .
- On  $X = \{0, 1\}$ ,  $\mathcal{T} = \{\emptyset, \{0\}, X\}$  is a topology not induced by any metric.

- $\mathcal{T}_1$  is **finer** than  $\mathcal{T}_2$  if  $\mathcal{T}_2 \subseteq \mathcal{T}_1$ ;  $\mathcal{T}_2$  is then coarser. Finer means more open sets. Two topologies need not be comparable.

The **subspace topology** on  $A \subseteq X$  is

$$\mathcal{T}_A = \{A \cap U : U \in \mathcal{T}\}.$$

This agrees with the topology of the restricted metric when  $X$  is a metric space.

**Convergence and neighbourhoods.** [III.7, III.12] A neighbourhood of  $x$  is an open set containing  $x$ . A sequence  $x_n$  converges to  $x$  if every neighbourhood of  $x$  contains all sufficiently late terms. In the trivial topology, every sequence converges to every point. In the discrete topology, convergence to  $x$  means eventual equality to  $x$ . Ex4 q3

**Hausdorff property** [III.15]. For any distinct  $x, y \in X$ , there exist disjoint open sets  $U, V$  with  $x \in U$  and  $y \in V$ . Ex5

**Main results** [III.17-III.18]. Every metric space is Hausdorff, and every sequence in a Hausdorff space has at most one limit. q6,8(i)

*Proof ideas.* In a metric space use the disjoint balls of radius  $d(x, y)/2$  about  $x$  and  $y$ . A sequence converging to both points would eventually belong to both disjoint neighbourhoods.

**Interior, limit points, closed sets and closure.** [III.11, III.20-III.24] The definitions use open sets rather than distances: Ex5  
q1,3,4

- $\text{int}(A)$  is the union of all open sets contained in  $A$ .
- $x \in A'$  if every open neighbourhood of  $x$  meets  $A \setminus \{x\}$ .
- $A$  is closed if  $A' \subseteq A$ , equivalently if  $X \setminus A$  is open.
- $\overline{A}$  is the intersection of all closed sets containing  $A$ . Equivalently, every neighbourhood of a point of  $\overline{A}$  meets  $A$ .

The usual union/intersection rules follow by taking complements. If a sequence in a closed set  $A$  converges in  $X$ , its limit lies in  $A$ .

**Keep in mind.** In arbitrary topological spaces, containing the limits of all convergent sequences from  $A$  need not make  $A$  closed. Likewise, a point of  $\overline{A}$  need not be obtainable as a sequential limit from  $A$ . The converses are valid for metric spaces; Hausdorffness alone does not supply them.

## 05 CONDENSED LECTURE NOTES

# Continuity, homeomorphisms and isometries

Source: IV, original pp. 20-23

**Continuity at a point in metric spaces.** [IV.1] For  $f : (X, \rho) \rightarrow (Y, d)$ , continuity at  $a$  means Ex6 q5

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in X : \rho(x, a) < \delta \implies d(f(x), f(a)) < \varepsilon.$$

Equivalently,  $f(B_\delta^o(a)) \subseteq B_\varepsilon^d(f(a))$ . The function is continuous if this holds at every point.

**Sequential criterion (Heine).** [IV.2] For maps between metric spaces,  $f$  is continuous at  $a$  if and only if  $x_n \rightarrow a$  always implies  $f(x_n) \rightarrow f(a)$ . Compositions of continuous maps are continuous [IV.3].

**Preimage.** [IV.7] For any map  $f : X \rightarrow Y$  and set  $A \subseteq Y$ ,

$$f^{-1}(A) = \{x \in X : f(x) \in A\}.$$

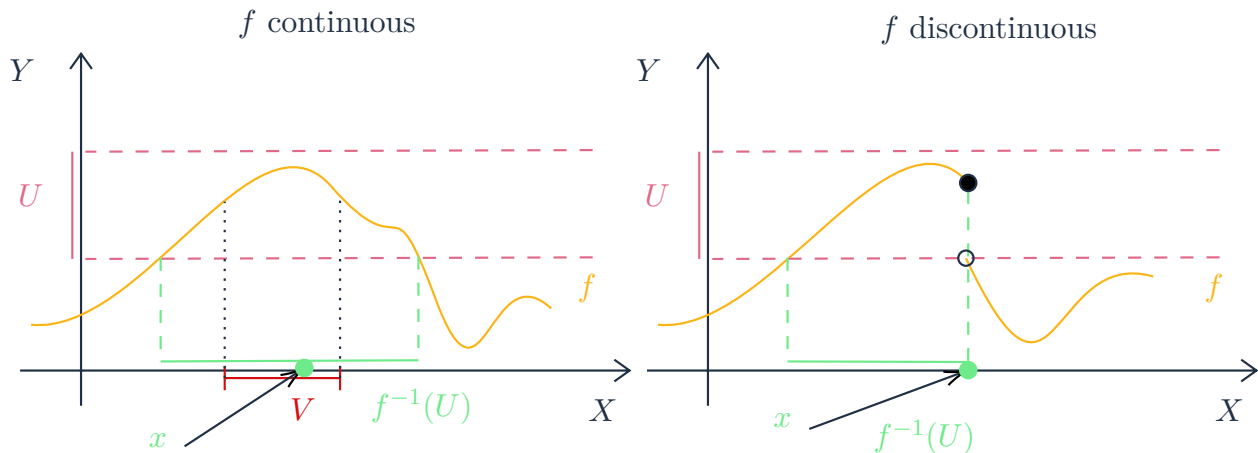
This notation does not require an inverse function. **Intuition:**  $f^{-1}(A)$  is the set of all points of  $X$  that get mapped by  $f$  into  $A$ .

**Topological characterization of continuity** [IV.8, IV.10]. The following are equivalent:

- (i)  $f$  is continuous;
- (ii)  $f^{-1}(U)$  is open in  $X$  for every open  $U \subseteq Y$ ;
- (iii)  $f^{-1}(F)$  is closed in  $X$  for every closed  $F \subseteq Y$ .

For arbitrary topological spaces, condition (ii) is the definition. Continuity concerns *preimages*, not images, of open sets.

**Visual intuition for the topological definition.** If  $x \in f^{-1}(U)$ , then  $f(x) \in U$ . For a continuous map we can find an open set  $V$  around  $x$  that stays inside  $f^{-1}(U)$ . For a discontinuous map this can fail: no open neighbourhood of  $x$  fits inside the preimage.



**Homeomorphism.** [IV.11] A homeomorphism  $f : X \rightarrow Y$  is a bijection such that both  $f$  and  $f^{-1}$  are continuous. Write  $X \cong Y$  when one exists. A continuous bijection alone is not enough. Ex6 q10,11

**Isometry.** [IV.12] A map  $f : (X, d) \rightarrow (Y, \rho)$  is an isometry if  $\rho(f(x), f(y)) = d(x, y)$  for all  $x, y$ . An isometry is injective and is a homeomorphism onto its image; a bijective isometry is an *isometric isomorphism*. A homeomorphism need not preserve distances. Ex6 q3,11

## 06 CONDENSED LECTURE NOTES

# Cauchy sequences, completeness and completion

Source: V.1.1-V.1.3, original pp. 24-27

**Cauchy sequence.** [V.1] A sequence  $(x_n)$  in  $(X, d)$  is Cauchy if

$$\forall \varepsilon > 0 \exists N_\varepsilon \forall m, n > N_\varepsilon : d(x_m, x_n) < \varepsilon.$$

Ex7  
q2,3

This condition compares late terms to one another; it does not assume that a limit already exists in  $X$ .

**Two fundamental lemmas.** [V.2-V.3]

- **Convergent  $\implies$  Cauchy.** Every convergent sequence is Cauchy.
- **Cauchy + convergent subsequence  $\implies$  convergence.** If a Cauchy sequence has a convergent subsequence with limit  $x \in X$ , then the whole sequence converges to  $x$ .

**Complete metric space.** [V.4] The space is complete if every Cauchy sequence converges to a point of that space. The spaces  $\mathbb{R}$  and  $\mathbb{C}$  with their standard metrics are complete. The interval  $(0, 1)$  is not:  $1/(n+1) \rightarrow 0 \notin (0, 1)$ . The rationals are not complete: rational approximations to  $\sqrt{2}$  form a Cauchy sequence with no rational limit. Ex7 q1,3

**Closed subspaces and completeness** [V.9-V.10].

- A closed subspace of a complete metric space is complete.
- A complete subspace of any metric space is closed in that ambient space.

Consequently, when  $X$  is complete,  $A \subseteq X$  is complete if and only if  $A$  is closed in  $X$ .

**Completion inside a complete space.** [V.13] If  $A \subseteq X$  and  $X$  is complete, then  $\overline{A}$  with the restricted metric is a completion of  $A$ . In particular, the completion of  $(0, 1)$  with its usual metric is  $[0, 1]$ .

**Complete function spaces.** [V.14-V.15] The space  $B(S)$  of bounded real- or complex-valued functions is complete in the supremum metric. So is  $C[a, b]$ , because a uniform limit of continuous functions is continuous. Equivalently,  $C[a, b]$  is a closed subspace of  $B([a, b])$ .

**Diameter.** [V.16] For nonempty  $A \subseteq X$ ,  $\text{diam}(A) = \sup\{d(x, y) : x, y \in A\}$ , possibly infinite.

**Cantor's intersection theorem** [V.17]. Let  $X$  be complete, and let  $A_1 \supseteq A_2 \supseteq \dots$  be nonempty closed subsets with  $\text{diam}(A_n) \rightarrow 0$ . Then

$$\bigcap_{n=1}^{\infty} A_n = \{x\} \quad \text{for exactly one } x \in X.$$

*Proof idea:* choose  $x_n \in A_n$ ; nesting and shrinking diameters make  $(x_n)$  Cauchy. Completeness supplies the limit and closedness puts it in every  $A_n$ .

**Keep in mind.** Shrinking diameters alone do not suffice: the nested sets  $(0, 1/n]$  in  $\mathbb{R}$  have empty intersection because they are not closed. Completeness is not a topological invariant:  $\mathbb{R} \cong (0, 1)$ , but their usual metrics have different completeness properties.

**07 CONDENSED LECTURE NOTES****Contractions and Picard's existence theorem**

Source: V.1.4-V.1.5, original pp. 27-30

**Lipschitz map and contraction.** [V.18, V.22] A map  $T : X \rightarrow X$  is a contraction if there is a constant  $0 \leq c < 1$  such that Ex7  
q5,7,8

$$d(Tx, Ty) \leq c d(x, y) \quad (x, y \in X).$$

A finite constant  $c \geq 0$  gives a Lipschitz condition; the strict inequality  $c < 1$  is essential to the contraction theorem. If a real function is continuously differentiable on an interval and  $\sup |T'| \leq c$ , the mean value theorem gives this bound.

**Banach fixed-point theorem** [V.19]. On a nonempty complete metric space, every contraction  $T : X \rightarrow X$  has a unique fixed point  $\eta$  satisfying  $T\eta = \eta$ . For every starting point  $x_0 \in X$ , the iterates

$$x_{n+1} = Tx_n, \quad x_n = T^n x_0$$

converge to  $\eta$ . **Error estimate** [V.20]:

$$d(x_n, \eta) \leq \frac{c^n}{1-c} d(x_0, Tx_0), \quad n \geq 0.$$

*Proof structure:* sum the geometric bounds on successive increments to show the iterates are Cauchy; use completeness; pass to the limit in  $x_{n+1} = Tx_n$ ; for uniqueness use  $d(\eta, \zeta) \leq c d(\eta, \zeta)$ .

**Example.** On  $\mathbb{R}$ ,  $T(x) = x/2 + 1$  has fixed point 2. Starting at 0,  $x_n = 2 - 2^{1-n}$  and the error is  $2^{1-n}$ . On the incomplete space  $(0, 1)$ ,  $T(x) = x/2$  is a contraction but has no fixed point in the space. If  $c = 1$ , the identity map has many fixed points.

**First-order initial-value problems**



Consider the autonomous initial-value problem

$$\varphi'(x) = f(\varphi(x)), \quad \varphi(x_0) = y_0.$$

**Picard's theorem.** [V.23] Assume  $f$  is continuous and Lipschitz:

$$|f(y_1) - f(y_2)| \leq L|y_1 - y_2|.$$

Then there is a unique local solution through  $(x_0, y_0)$  on some interval around  $x_0$ .

This comes from Banach's fixed-point theorem. The differential equation is equivalent to the integral equation

$$\varphi(x) = y_0 + \int_{x_0}^x f(\varphi(t)) dt,$$

and on a sufficiently short interval the associated integral operator

$$(T\varphi)(x) = y_0 + \int_{x_0}^x f(\varphi(t)) dt$$

is a contraction on a suitable complete function space. Hence  $T$  has a unique fixed point, which is exactly the required solution. This gives a *local* existence-and-uniqueness theorem.

## 08 CONDENSED LECTURE NOTES

# Connectedness and path-connectedness

Source: V.2, original pp. 31-34

**Separation and connectedness.** [V.24] A separation of  $X$  is a representation  $X = A \cup B$  with  $A, B$  disjoint and open in  $X$ . It is nontrivial if both are nonempty. The space is connected if it has no nontrivial separation. A subset is connected when given its subspace topology.

**Clopen sets.** [V.26-V.27] A set is *clopen* if it is both open and closed. The space  $X$  is connected if and only if its only clopen subsets are  $\emptyset$  and  $X$ . Ex8 q3

**Connected subsets of the real line.** [V.28-V.30] A set  $I \subseteq \mathbb{R}$  is an interval if  $x, z \in I$  and  $x < y < z$  imply  $y \in I$ . A subset of  $\mathbb{R}$  with the standard topology is connected if and only if it is an interval, including degenerate intervals. Thus  $\mathbb{R}$  and all usual intervals are connected;  $(0, 1) \cup (2, 3)$  is disconnected. Ex9 q3(ii)

**Characteristic function criterion.** [V.31-V.34] For  $E \subseteq X$ , let  $\chi_E$  take the value 1 on  $E$  and 0 elsewhere. Then  $E$  is clopen if and only if  $\chi_E : X \rightarrow \mathbb{R}$  is continuous. Consequently,

$$X \text{ connected} \iff \text{every continuous } f : X \rightarrow \{0, 1\} \text{ is constant,}$$

where  $\{0, 1\}$  has its discrete topology.

**Path and path-connectedness.** [V.35-V.36] A path from  $x$  to  $y$  is a continuous map  $\gamma : [0, 1] \rightarrow X$  with  $\gamma(0) = x$  and  $\gamma(1) = y$ . The space is path-connected if every pair of points can be joined by a path. A subset  $S \subseteq X$  is path-connected exactly when the joining paths can be chosen in  $X$  with their images entirely in  $S$ . Ex9 q1,2,3

**Concatenating paths.** If  $\gamma_{12}(1) = \gamma_{23}(0)$ , one can traverse them consecutively:

$$\gamma(t) = \begin{cases} \gamma_{12}(2t), & 0 \leq t \leq 1/2, \\ \gamma_{23}(2t - 1), & 1/2 \leq t \leq 1. \end{cases}$$

### Main implications [V.39-V.40].

- Every path-connected space is connected.
- A continuous image of a connected space is connected.

The second statement includes the intermediate value theorem: a continuous image of a real interval is an interval. Connectedness is therefore preserved by homeomorphisms.

**Connected, but not path-connected.** [V.38] The precise sine-curve example in the notes is

$$S = \{(x, \sin(1/x)) : 0 < x \leq 1\} \cup \{(0, 0)\}.$$

It is connected because the graph is connected and the added origin is in its closure. There is no path in  $S$  joining the origin to the graph. Only the origin, not an entire vertical segment, is added in this example.

**Using connectedness to rule out homeomorphisms** [V.42-V.44]

- $\mathbb{R} \not\cong (0, 1) \cup (1, 2)$ , because the target is disconnected.
- $\mathbb{R}^2 \setminus \{0\}$  is path-connected: use a straight segment if possible, or a two-segment path avoiding the origin.
- $\mathbb{R} \not\cong \mathbb{R}^2$ . A hypothetical homeomorphism would identify the punctured plane with the real line minus one point; the former is connected and the latter is not.

## 09 CONDENSED LECTURE NOTES

# Compactness: covers, sequences and finite nets

Source: V.3.1-V.3.3, original pp. 35-40

**Open cover and compactness.** [V.50] An open cover of  $X$  is a family  $(U_i)_{i \in I}$  of open subsets with  $X = \bigcup_i U_i$ . The space is compact if every open cover has a finite subcover. For  $K \subseteq X$ , this means every family of ambient open sets covering  $K$  has a finite subfamily still covering  $K$ .

**Sequential compactness.** [V.45-V.46] A metric space is sequentially compact if every sequence has a subsequence converging to a point *in the space*. Bolzano-Weierstrass gives this property for  $[a, b] \subset \mathbb{R}$ . A sequence need not itself converge.

**Finite intersection property (FIP).** [V.51-V.53] A family  $(C_i)$  has the FIP if every finite subfamily has nonempty intersection.

$X$  compact  $\iff$  every family of closed subsets of  $X$  with the FIP has nonempty total intersection.

This criterion is the open-cover definition rewritten using complements.

**Boundedness and total boundedness.** [V.57-V.60] A metric space is bounded if it is contained in a ball of finite radius, equivalently if its diameter is finite. It is *totally bounded* if

$$\forall \varepsilon > 0 \exists x_1, \dots, x_N \in X : \quad X = \bigcup_{j=1}^N B_\varepsilon(x_j).$$

The finite set of centres may depend on  $\varepsilon$ . Total boundedness implies boundedness; the converse fails in general.

**The central compactness theorem for metric spaces** [V.61]:

$$\text{compact} \iff \text{sequentially compact} \iff \text{complete and totally bounded.}$$

In particular, sequential compactness implies completeness [V.49]. The displayed equivalences are metric-space results; they must not be assumed for arbitrary topological spaces.

**Closedness and boundedness of compact sets.** [V.55-V.59]

- A closed subset of a compact topological space is compact.
- A compact subset of a Hausdorff space is closed.
- A compact metric space is bounded.

Thus a compact subset of a metric space is closed and bounded, but those two conditions alone are not sufficient in a general metric space.

**Heine-Borel theorem.** [V.64] For  $K \subseteq \mathbb{R}^n$  with the Euclidean topology,

$$K \text{ compact} \iff K \text{ is closed and bounded.}$$

### Examples and counterexamples

- $\mathbb{R}$  is not compact:  $\{(-n, n) : n \geq 1\}$  covers it, but no finite subfamily does. Equivalently,  $[n, \infty)$  have the FIP and empty total intersection.
- $(0, 1)$  is not compact:  $1/(n+1)$  has no subsequence converging in  $(0, 1)$ . The open cover  $\{(1/n, 1) : n \geq 2\}$  has no finite subcover.
- An infinite discrete metric space has diameter 1 and is complete, but is not totally bounded: balls of radius  $1/2$  are singletons. It is closed in itself and bounded, yet not compact.
- $(0, 1)$  is totally bounded but not complete;  $\mathbb{R}$  is complete but not totally bounded. Both hypotheses in the metric criterion are needed.

## 10 CONDENSED LECTURE NOTES

# Compactness and continuous functions

Source: V.3.4-V.3.5, original pp. 41-42

**Continuous images of compact spaces.** [V.65] If  $X$  is compact and  $f : X \rightarrow Y$  is continuous, then  $f(X)$  is compact, with its subspace topology. No Hausdorff assumption on  $Y$  is needed for this statement. Ex10 q4

*Proof idea.* Pull an open cover of  $f(X)$  back to an open cover of  $X$ , choose a finite subcover there, then use its corresponding sets to cover the image.

**Immediate consequence for real functions.** A continuous real-valued function on a nonempty compact space is bounded and attains its maximum and minimum, because its image is a nonempty compact subset of  $\mathbb{R}$ .

**Compact-to-Hausdorff bijection theorem** [V.67]. If  $X$  is compact,  $Y$  is Hausdorff, and  $f : X \rightarrow Y$  is a continuous bijection, then  $f^{-1}$  is continuous. Hence  $f$  is a homeomorphism.

*Reason.* A closed subset of  $X$  is compact; its image is compact and therefore closed in  $Y$ . So  $f$  is a closed bijection, which makes its inverse continuous.

**A continuous bijection need not be a homeomorphism.** [V.68] On  $C^1[0, 1]$ , compare

$$d(f, g) = \|f - g\|_\infty, \quad \rho(f, g) = \|f - g\|_\infty + \|f' - g'\|_\infty.$$

The identity map  $(C^1, \rho) \rightarrow (C^1, d)$  is a continuous bijection. Its inverse is not continuous: for  $f_n(x) = \sin(n^2 x)/n$ ,

$$d(f_n, 0) \leq 1/n \rightarrow 0, \quad \rho(f_n, 0) \geq |f'_n(0)| = n.$$

Thus the two topologies differ. The compact-domain hypothesis in the preceding theorem cannot simply be discarded.

**Uniform continuity.** [V.69] A function  $f : (X, d) \rightarrow \mathbb{R}$  or  $\mathbb{C}$  is uniformly continuous if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in X : d(x, y) < \delta \implies |f(x) - f(y)| < \varepsilon.$$

The same  $\delta$  must work throughout  $X$ . In ordinary continuity,  $\delta$  may also depend on the point about which continuity is being tested. Uniform continuity implies continuity.

**Cantor's theorem / Heine-Cantor theorem** [V.70]. Every continuous real- or complex-valued function on a compact metric space is uniformly continuous.

*Proof idea.* Use continuity with output tolerance  $\varepsilon/2$ , cover  $X$  by smaller neighbourhoods, select finitely many by compactness, and take a positive minimum of their radii. The triangle inequality supplies the global estimate.

### A practical comparison

| Function and domain           | Uniform-continuity test                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------|
| $f(x) = x^2$ on $[0, 1]$      | $ x^2 - y^2  \leq 2 x - y $ ; choose $\delta = \varepsilon/2$ .                                  |
| $f(x) = x^2$ on $[0, \infty)$ | Take $x_n = n$ , $y_n = n + 1/n$ . The input gap tends to 0, but the output gap is $2 + 1/n^2$ . |
| Any Lipschitz function        | A fixed Lipschitz constant gives a global $\delta$ ; compactness is not necessary.               |

**Keep in mind.** Compactness is a topological invariant: since  $[0, 1]$  is compact and  $\mathbb{R}$  is not,  $[0, 1] \not\cong \mathbb{R}$  [V.66]. Uniform continuity is a metric notion; mere continuity on a noncompact domain does not guarantee it.

## 11 CONDENSED LECTURE NOTES

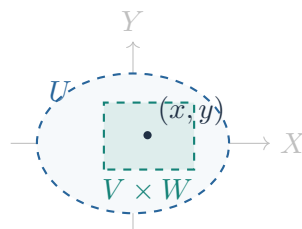
# Product spaces and coordinate maps

Source: VI.1, original pp. 43-46; also V.62-V.63

**Product topology.** [VI.1-VI.4] For topological spaces  $X, Y$ , a subset  $U \subseteq X \times Y$  is open if every  $(x, y) \in U$  has open neighbourhoods  $V \subseteq X$  and  $W \subseteq Y$  such that

$$(x, y) \in V \times W \subseteq U.$$

The sets  $V \times W$  form a *basis*: arbitrary unions of these basic rectangles are exactly the open sets. An open set need not itself be one rectangle.



Locally, an open set contains a product of neighbourhoods. The shape of the full open set is unrestricted.

**Projection maps and their defining property.** [VI.6-VI.7] The maps  $\pi_X(x, y) = x$  and  $\pi_Y(x, y) = y$  are continuous because

$$\pi_X^{-1}(V) = V \times Y, \quad \pi_Y^{-1}(W) = X \times W.$$

The product topology is the *coarsest* topology on  $X \times Y$  making both projections continuous. Intersections of these two types of strips give the basic rectangles.

**Coordinatewise continuity** [VI.13]. For a topological space  $T$  and a map  $g : T \rightarrow X \times Y$ ,

$$g \text{ continuous} \iff \pi_X \circ g \text{ and } \pi_Y \circ g \text{ are continuous.}$$

This concerns a map *into* a product. It does not say that a map *out of* a product is continuous merely because it is separately continuous.

**Metric products.** [VI.9-VI.10; IV.2 (equation)] For metric spaces  $(X, d_X)$  and  $(Y, d_Y)$ , the metric

Ex1 q6

$$\sigma((x_1, y_1), (x_2, y_2)) = \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\}$$

induces the product topology. Its balls are exactly

$$B_\varepsilon^{X \times Y}((x, y)) = B_\varepsilon^X(x) \times B_\varepsilon^Y(y).$$

The metric  $\sqrt{d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2}$  induces the same topology, since it lies between  $\sigma$  and  $\sqrt{2}\sigma$ .

**Coordinatewise convergence.** [VI.11] In the product topology,  $(x_n, y_n) \rightarrow (x, y)$  if and only if  $x_n \rightarrow x$  in  $X$  and  $y_n \rightarrow y$  in  $Y$ . This holds for arbitrary topological spaces, not only metric ones.

### Properties preserved by finite products

- Hausdorff  $X, Y$  imply Hausdorff  $X \times Y$  [VI.8].
- Connected  $X, Y$  imply connected  $X \times Y$  [VI.12].
- Compact metric  $X, Y$  have compact product [V.62]; the finite-product statement also holds for general compact spaces.

## 12 CONDENSED LECTURE NOTES

# Quotient spaces and identifications

Source: VI.2, original pp. 46-49

**Equivalence classes and quotient map.** [VI.14] For an equivalence relation  $\sim$  on  $X$ , write  $[x] = \{y \in X : y \sim x\}$  and let  $X/\sim$  be the set of equivalence classes. The canonical surjection is

$$p : X \longrightarrow X/\sim, \quad p(x) = [x].$$

The **quotient topology** is defined by

$$U \subseteq X/\sim \text{ is open } \iff p^{-1}(U) \text{ is open in } X.$$

It is the *finest* topology on the quotient making  $p$  continuous. A set of the form  $p^{-1}(U)$  is saturated: it is a union of whole equivalence classes.

**Topology induced by a map.** [VI.16] More generally, for  $f : X \rightarrow Y$  with  $Y$  initially only a set, declare  $U \subseteq Y$  open precisely when  $f^{-1}(U)$  is open in  $X$ . This is the finest topology making  $f$  continuous.

**Universal property** [VI.17]. When  $Y$  has the topology just induced by  $f$ , a map  $g : Y \rightarrow Z$  is continuous if and only if  $g \circ f : X \rightarrow Z$  is continuous.

For a quotient, a continuous map  $h : X \rightarrow Z$  which is constant on equivalence classes therefore descends to a unique continuous map  $\tilde{h} : X/\sim \rightarrow Z$ , satisfying  $h = \tilde{h} \circ p$ .

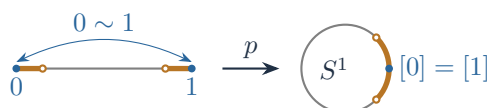
**Preserved properties.** [VI.15] A quotient of a connected space is connected; a quotient of a compact space is compact. Both follow because  $p$  is continuous and surjective. A quotient of a Hausdorff space need not be Hausdorff.

### The circle: identify endpoints [VI.18]

Identifying  $0 \sim 1$  in  $[0, 1]$  gives a circle. Equivalently, on  $\mathbb{R}$  use  $x \sim y$  if  $x - y \in \mathbb{Z}$ ; then

$$\mathbb{R}/\mathbb{Z} \cong S^1, \quad [x] \mapsto e^{2\pi i x}.$$

The map is well-defined and bijective. It is continuous by the universal property. Its domain is compact, as the image of  $[0, 1]$  under the quotient map, and  $S^1$  is Hausdorff; hence the bijection is a homeomorphism.



An open arc about the glued point pulls back to  $[0, \varepsilon) \cup (1 - \varepsilon, 1]$ , which is open in the subspace  $[0, 1]$ .

### The torus and a non-Hausdorff quotient [VI.19-VI.20]

- $\mathbb{R}^2/\mathbb{Z}^2 \cong \mathbb{T}^2 \cong S^1 \times S^1$ , with  $x \sim y$  when  $x - y \in \mathbb{Z}^2$ . Equivalently, glue opposite sides of  $[0, 1]^2$ :  $(0, t) \sim (1, t)$  and  $(s, 0) \sim (s, 1)$ . All four corners become one point.
- $\mathbb{R}/\mathbb{Q}$ , with  $x \sim y$  when  $x - y \in \mathbb{Q}$ , has the trivial topology. Every nonempty saturated open subset of  $\mathbb{R}$  is all of  $\mathbb{R}$ , since rational translates of any nonempty interval cover the line.